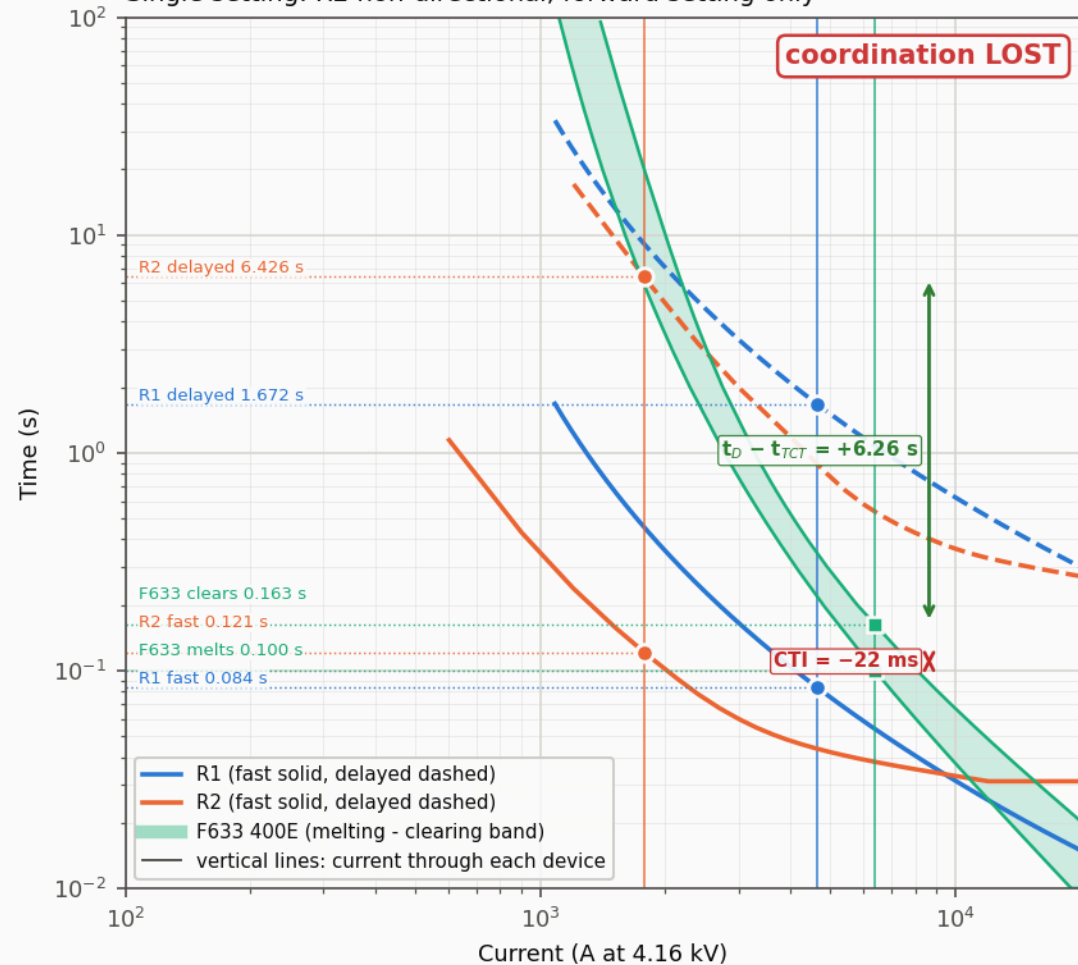
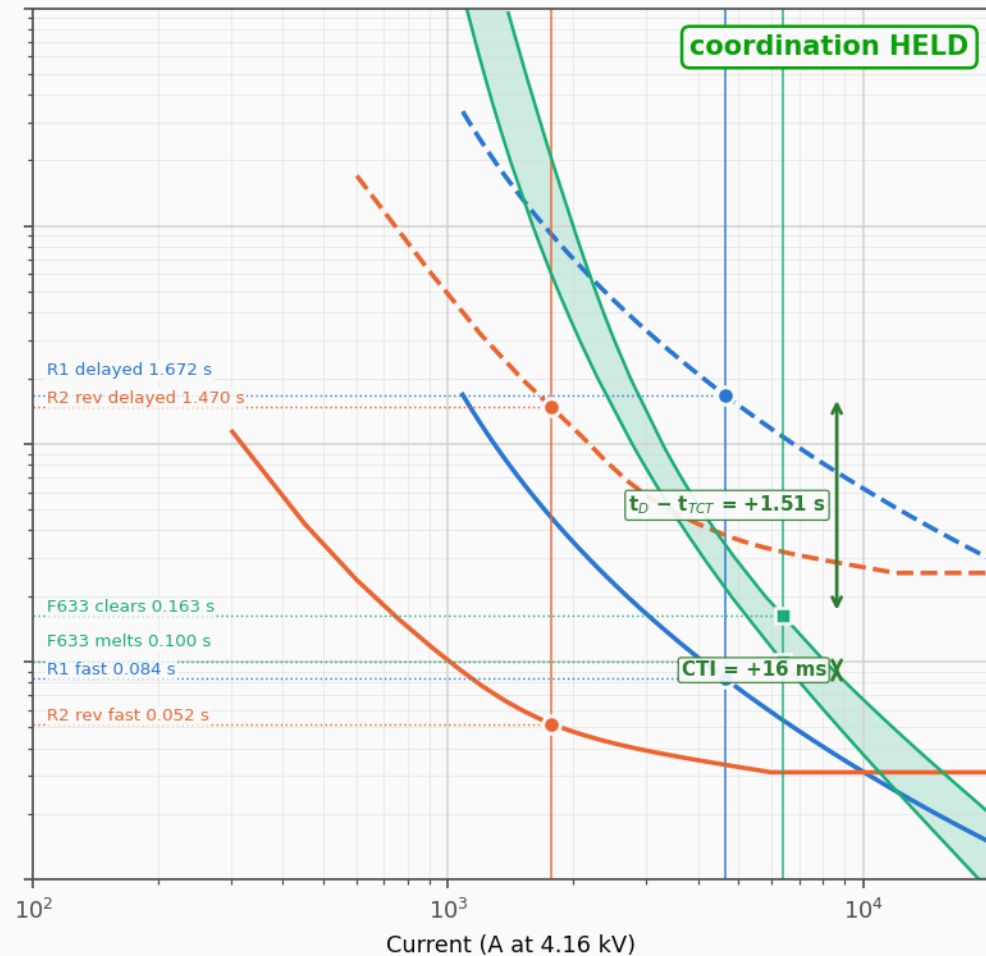


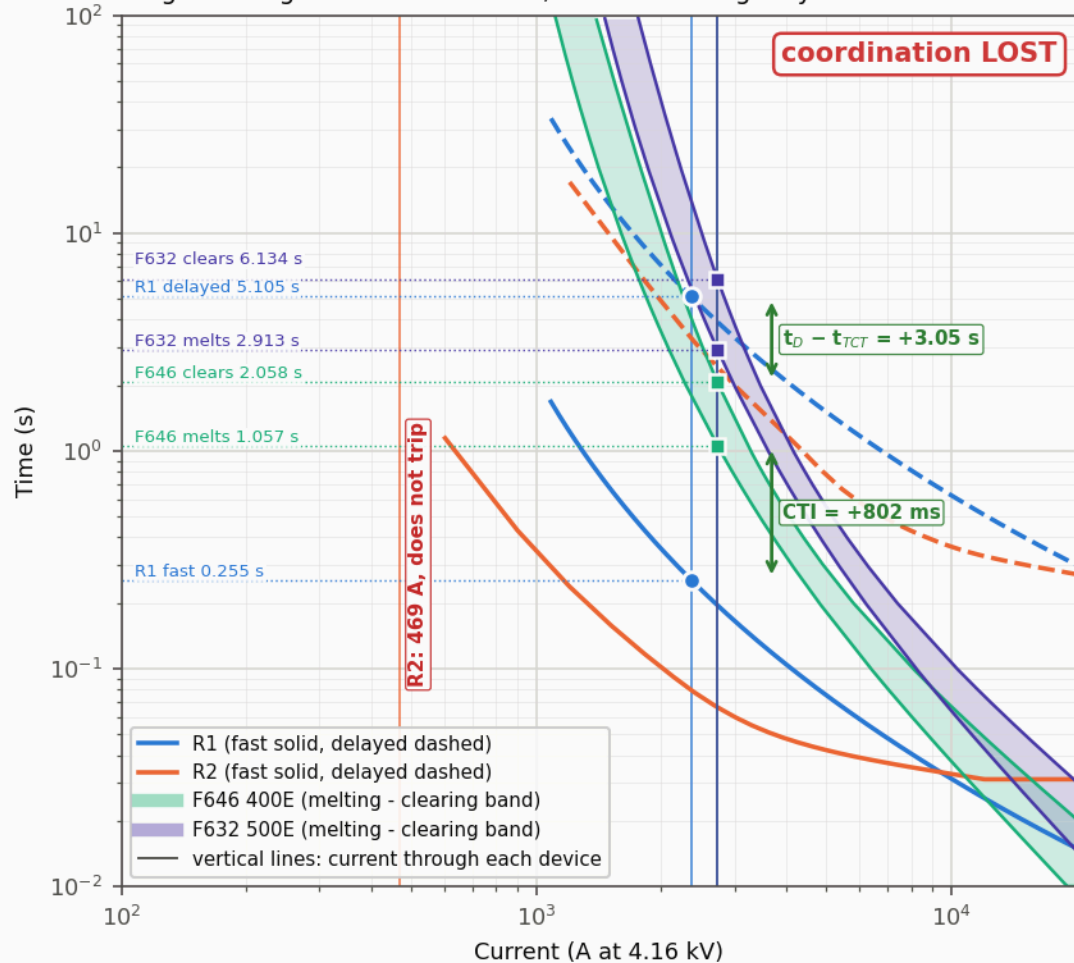
Single setting: R2 non-directional, forward setting only



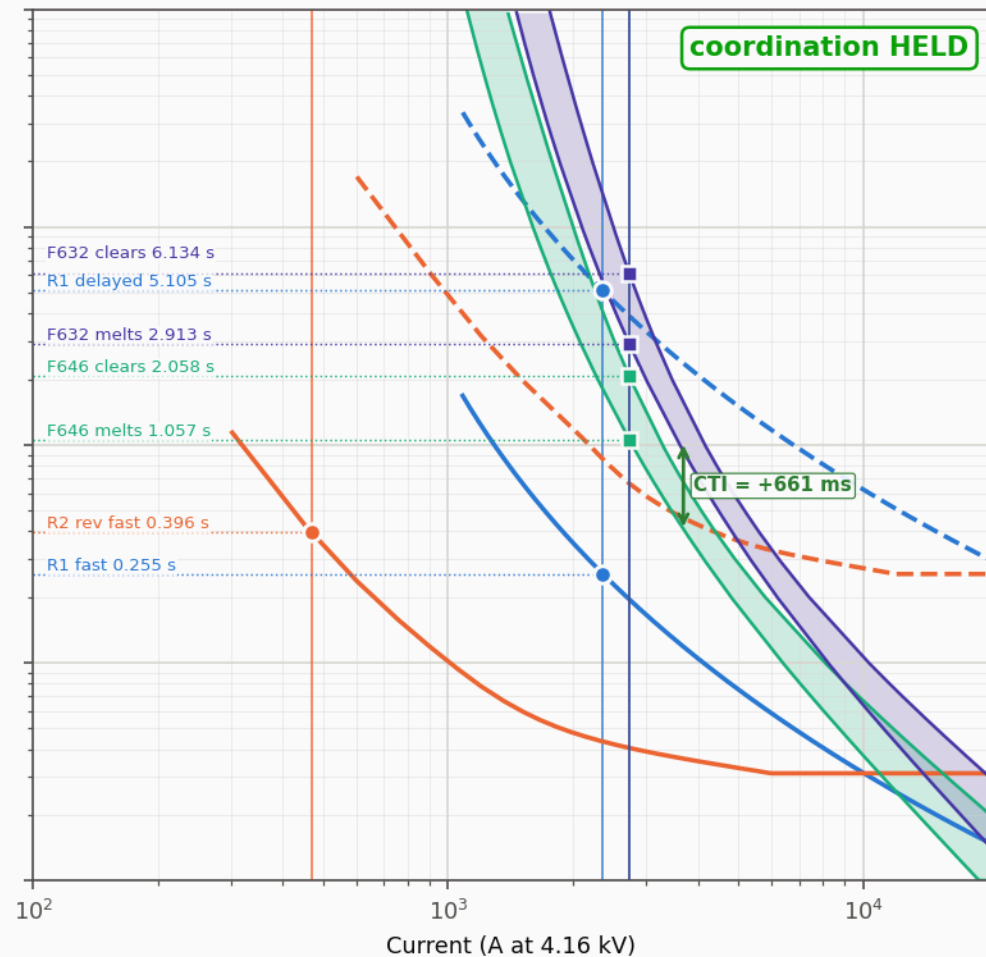
Dual setting: R2 as DSDR



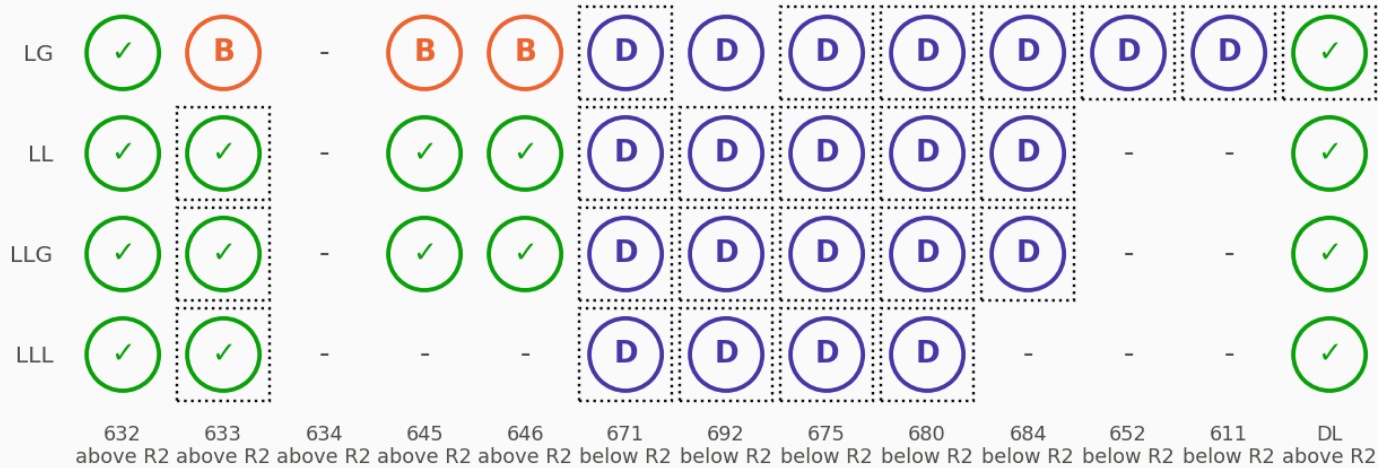
Single setting: R2 non-directional, forward setting only



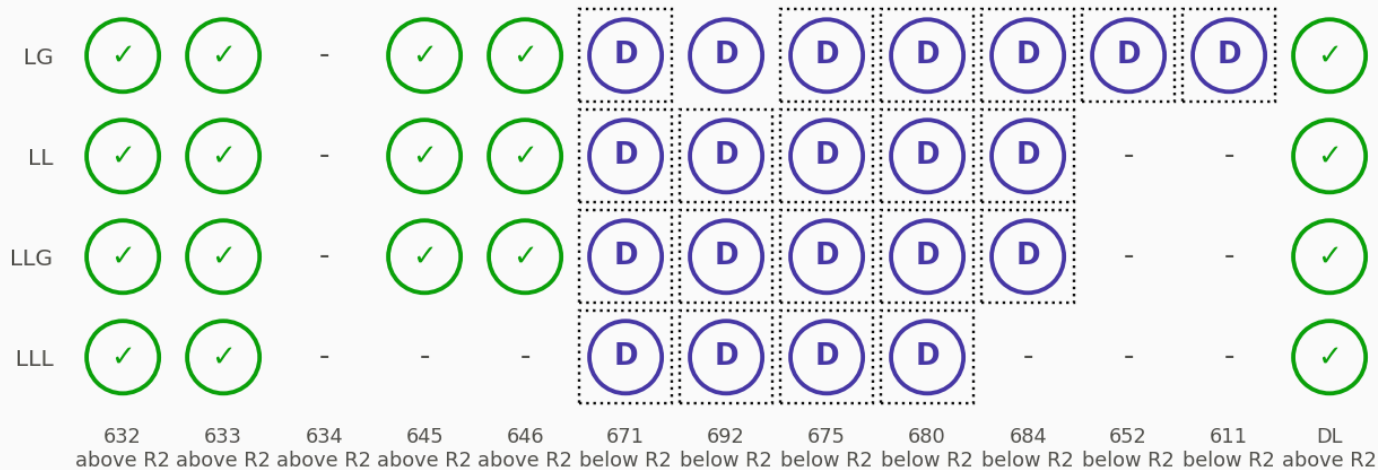
Dual setting: R2 as DSDR



Single setting on R2 - time sequence: above R2 15 of 18 held; below R2 the DG keeps feeding 21 of 21



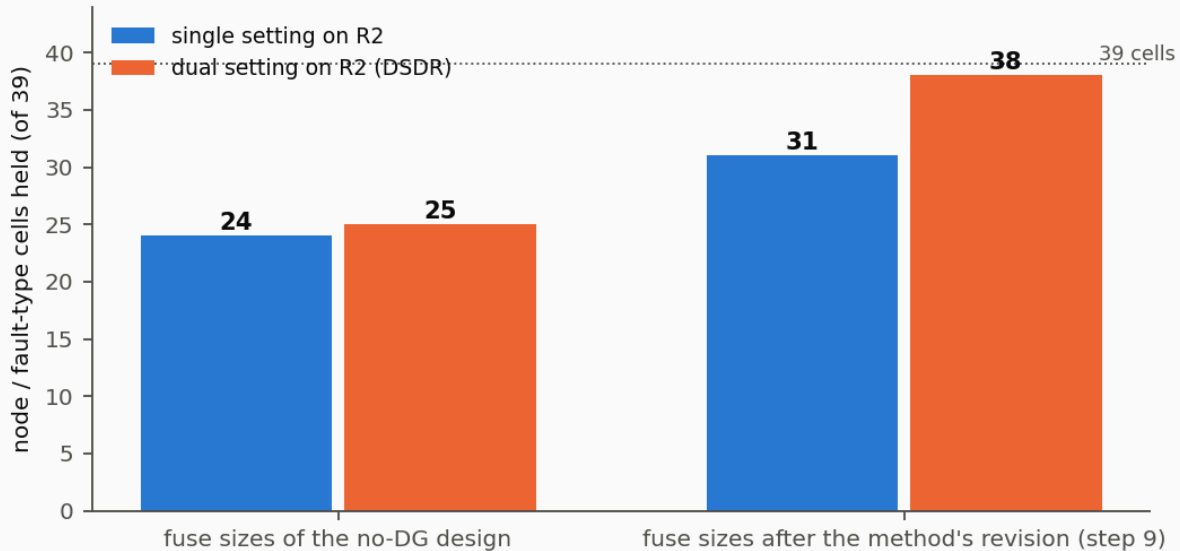
Dual setting on R2 (DSDR) - time sequence: above R2 18 of 18 held; below R2 the DG keeps feeding 21 of 21



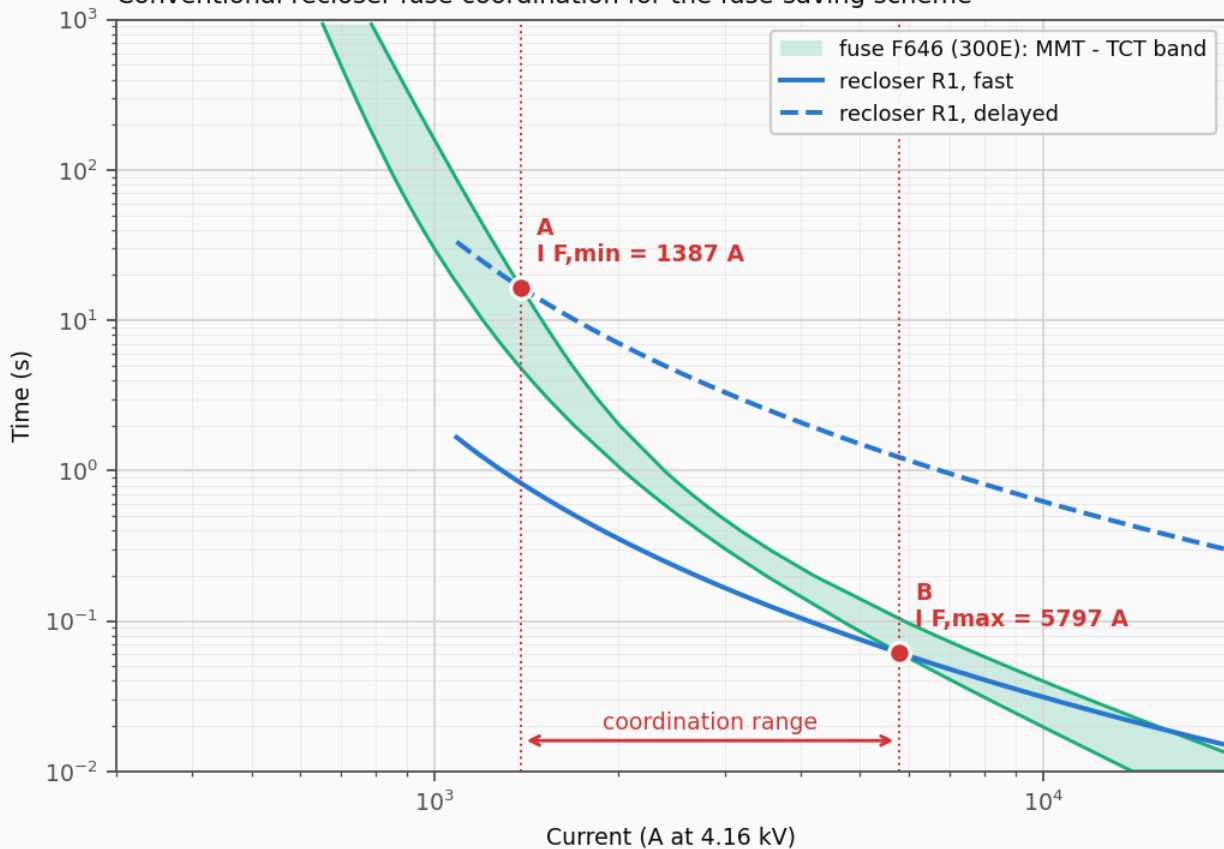
✓ held: fault cut off, fuse below 100 % heat  
 ○ fast sequence fine, a delayed-trip / series condition fails  
 D below R2: the DG keeps feeding the fault

B R2 does not trip on the DG's reverse current  
 X fuse melts before the reclosers finish  
 ... differs from the strict rule

## Fuse-saving coordination with the DG connected



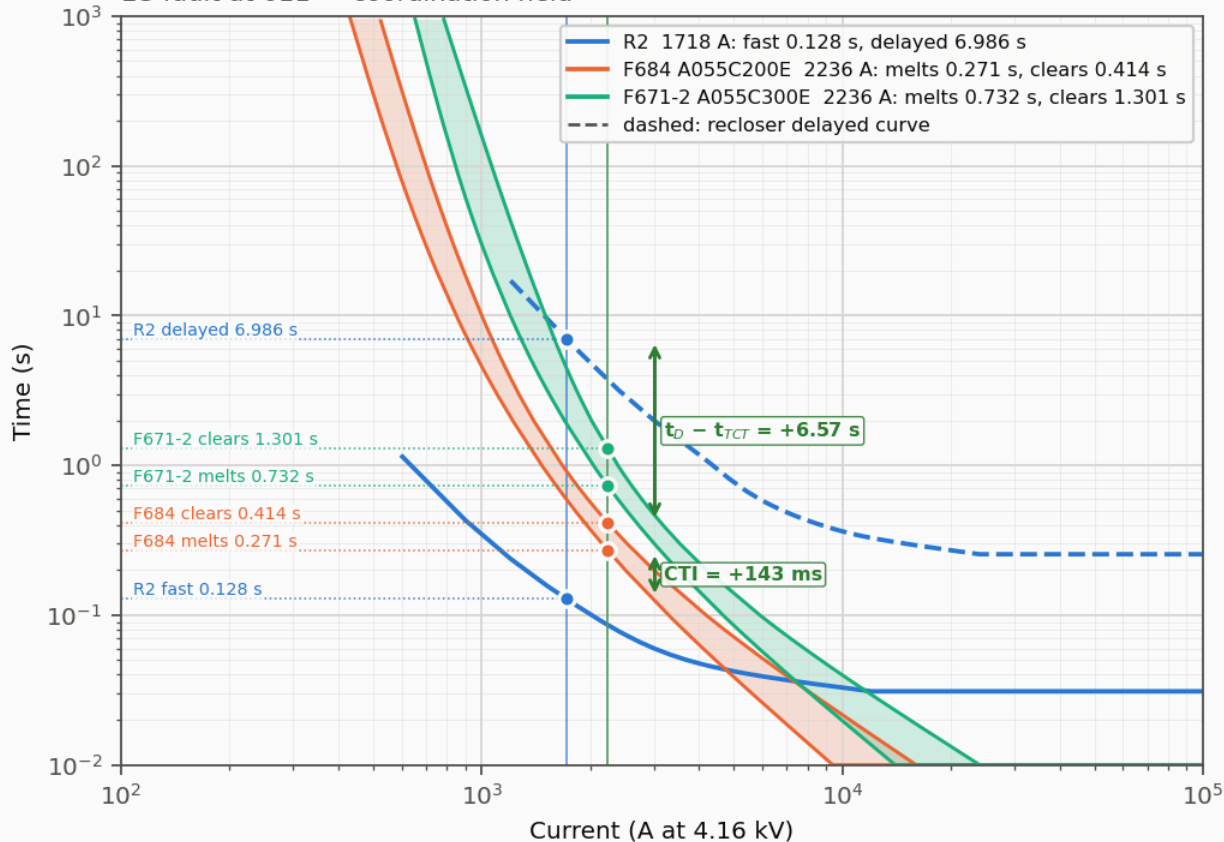
Conventional recloser-fuse coordination for the fuse-saving scheme



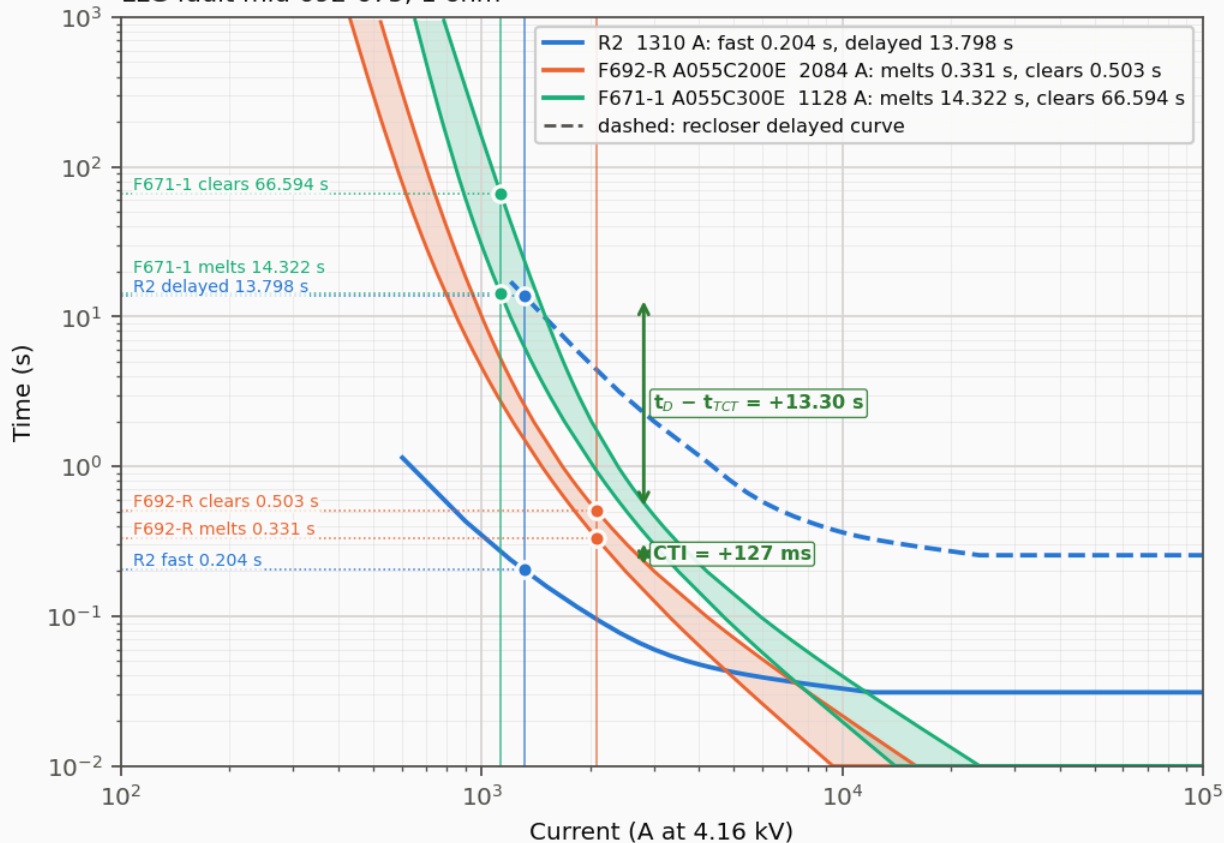
Time-current characteristics of the dual-setting recloser R2: reverse and forward setting



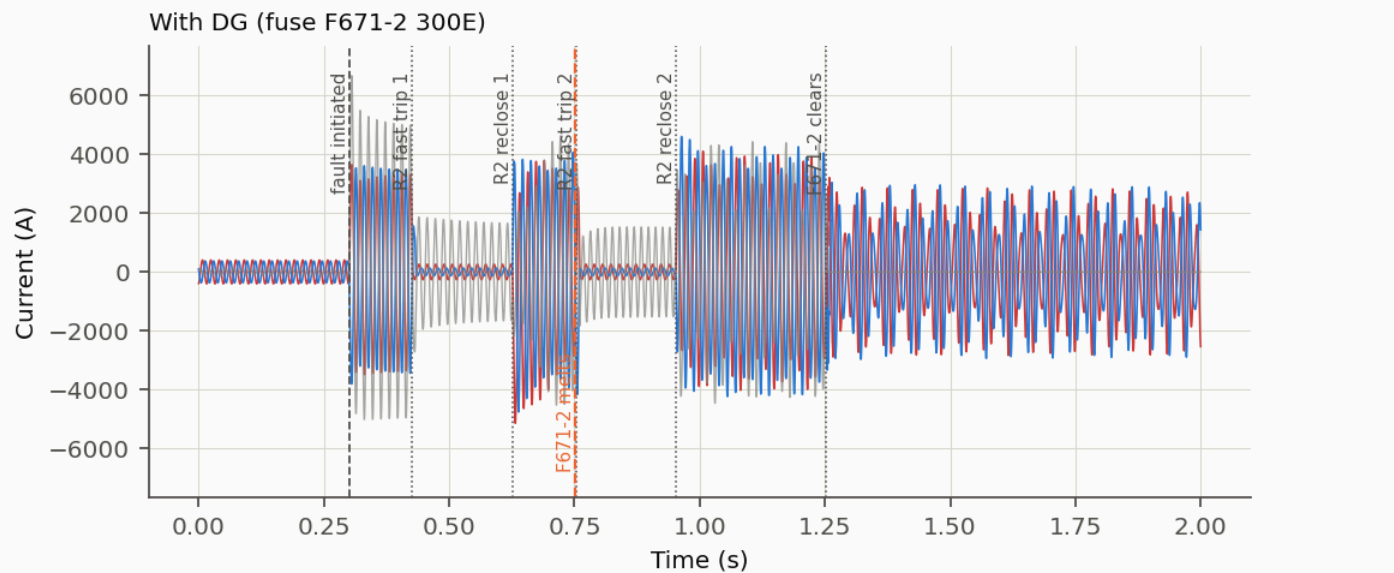
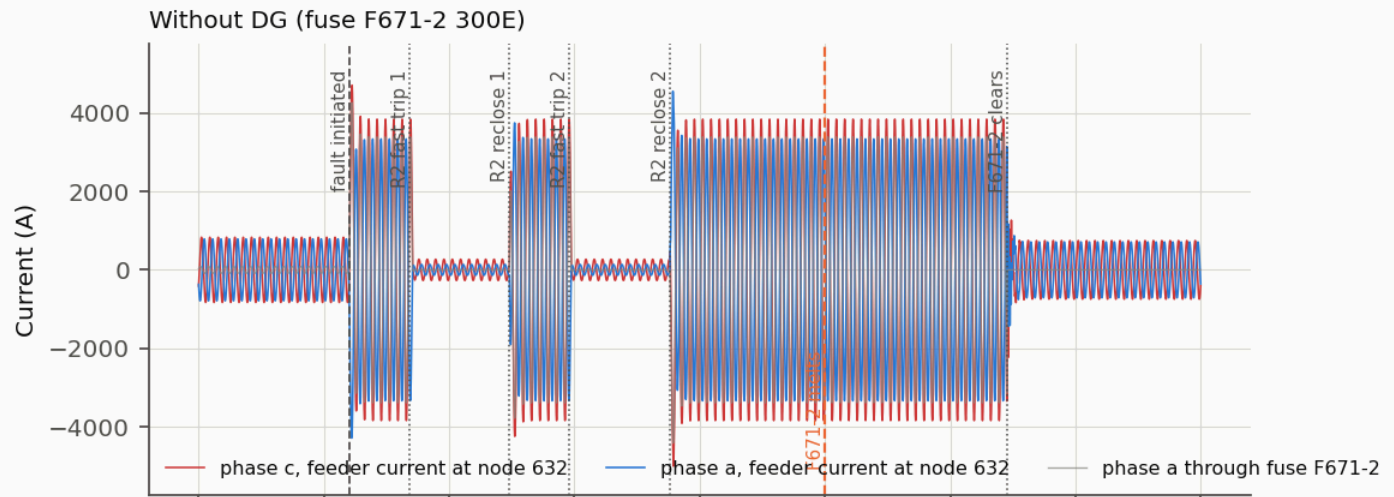
# LG fault at 611 - coordination held



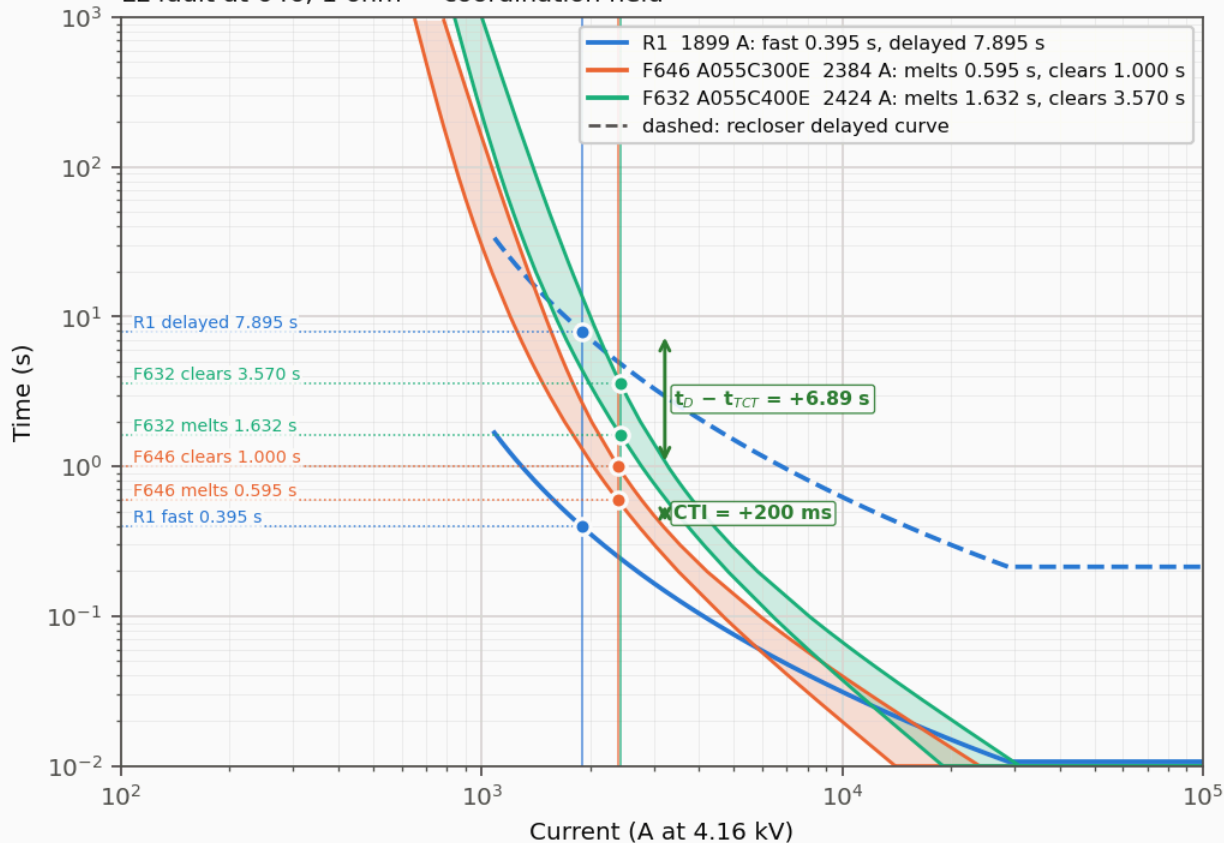
# LLG fault mid 692-675, 1 ohm



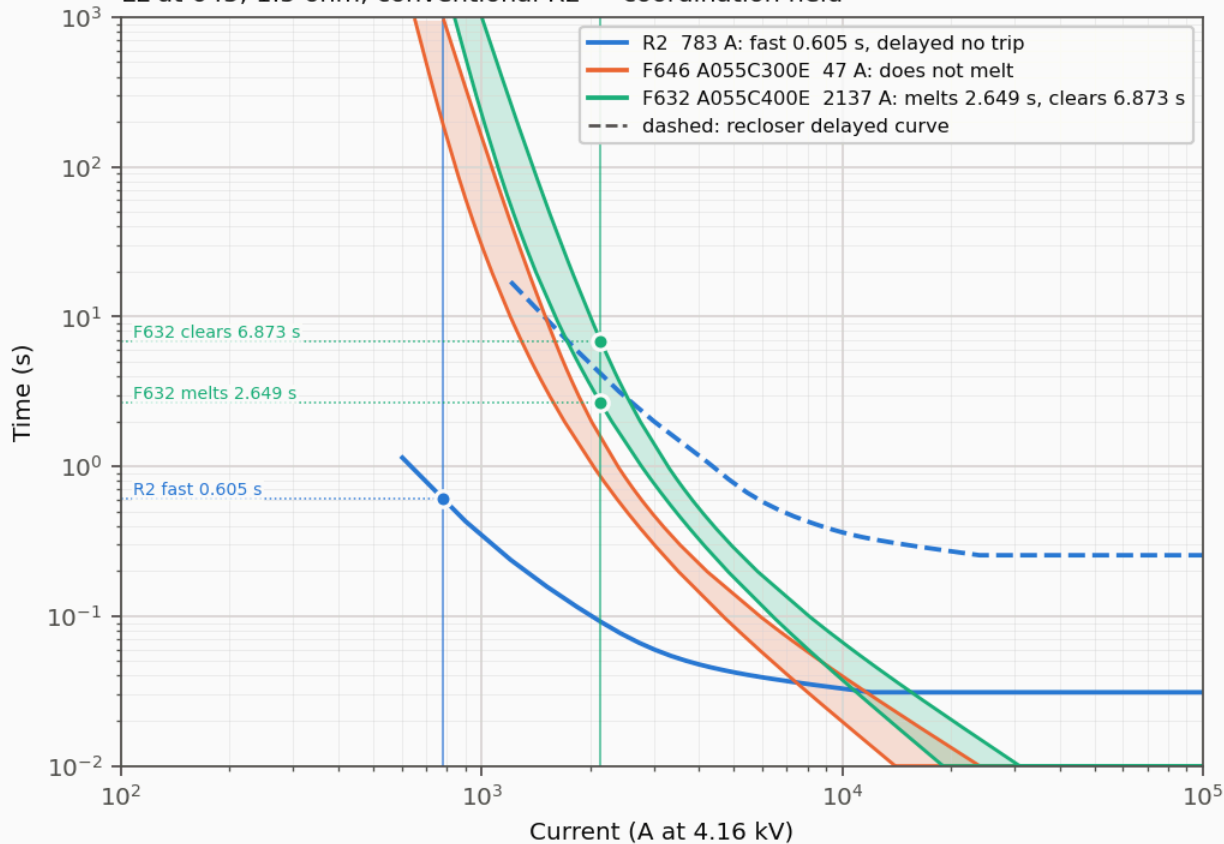
Current at node 632, LL a-c fault at 684 through 0.2 ohm, PowerFactory EMT



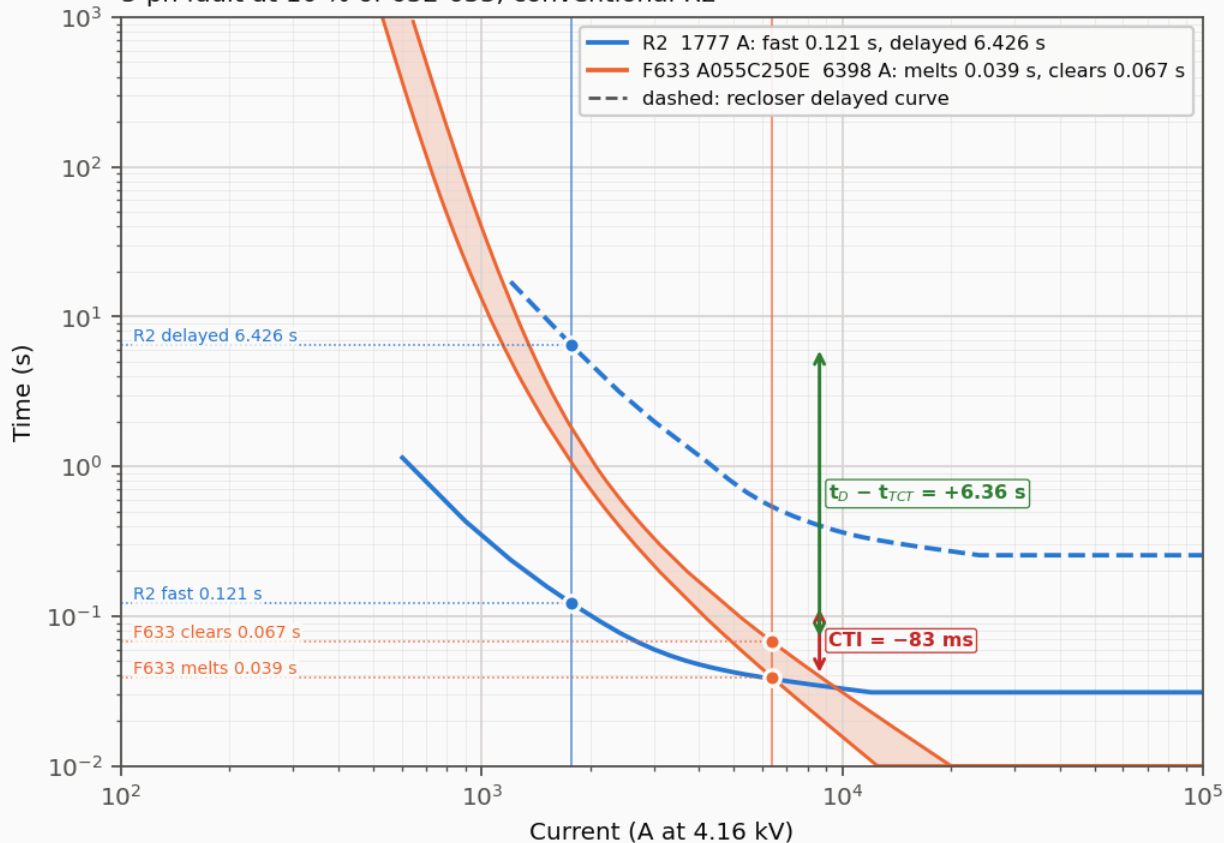
# LL fault at 646, 1 ohm - coordination held



# LL at 645, 1.5 ohm, conventional R2 - coordination held



### 3-ph fault at 10 % of 632-633, conventional R2



# Coordination without DSDR, DG 4.05 MVA at 692

|     |                                                                                   |                                                                                   |     |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                    |                                                                                    |                                                                                     |
|-----|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| LG  |   |   | —   |   |   |   |   |   |   |   |  |  |   |
| LL  |  |  | —   |  |  |  |  |  |  |  | —                                                                                  | —                                                                                  |  |
| LLG |  |  | —   |  |  |  |  |  |  |  | —                                                                                  | —                                                                                  |  |
| LLL |  |  | —   | —                                                                                  | —                                                                                   |  |  |  |  | —                                                                                   | —                                                                                  | —                                                                                  |  |
|     | 632                                                                               | 633                                                                               | 634 | 645                                                                                | 646                                                                                 | 671                                                                                 | 692                                                                                 | 675                                                                                 | 680                                                                                 | 684                                                                                 | 652                                                                                | 611                                                                                | DL                                                                                  |

Faulted node



coordination held

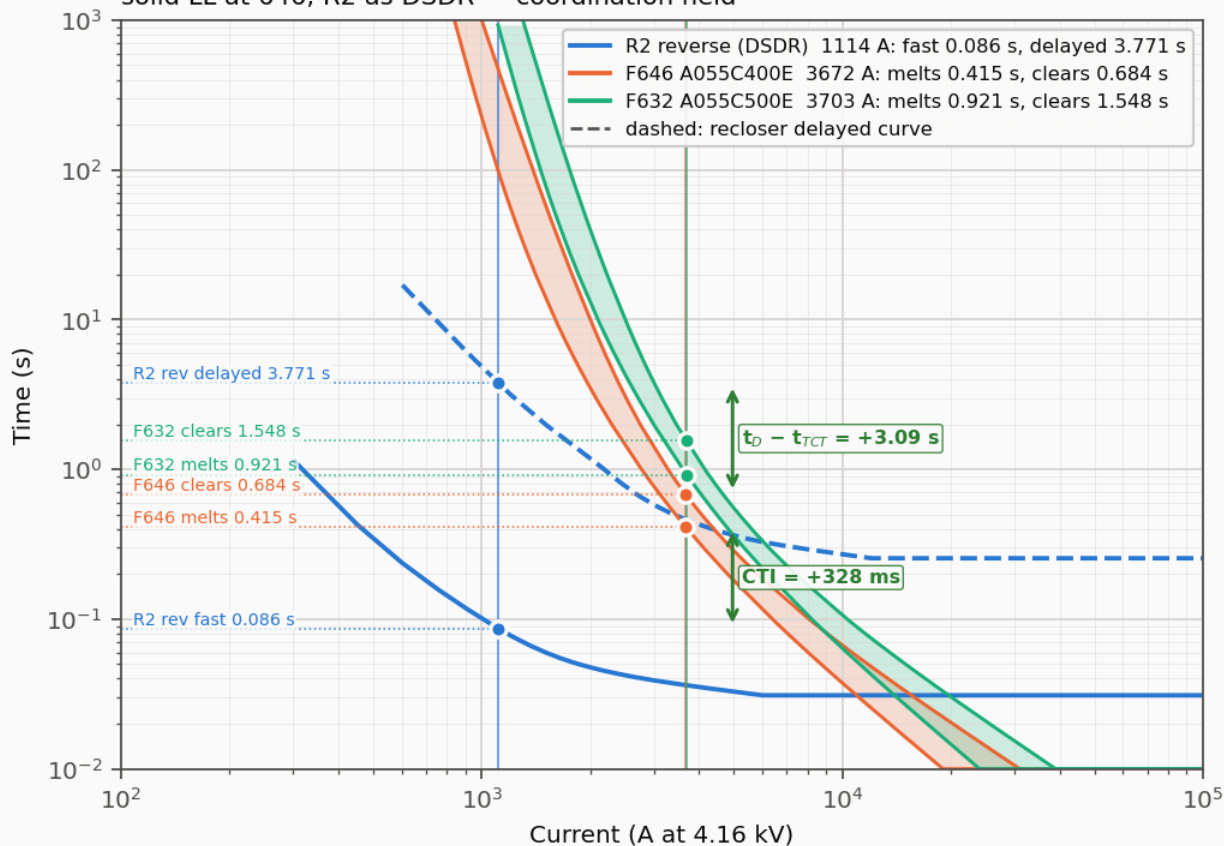


coordination lost

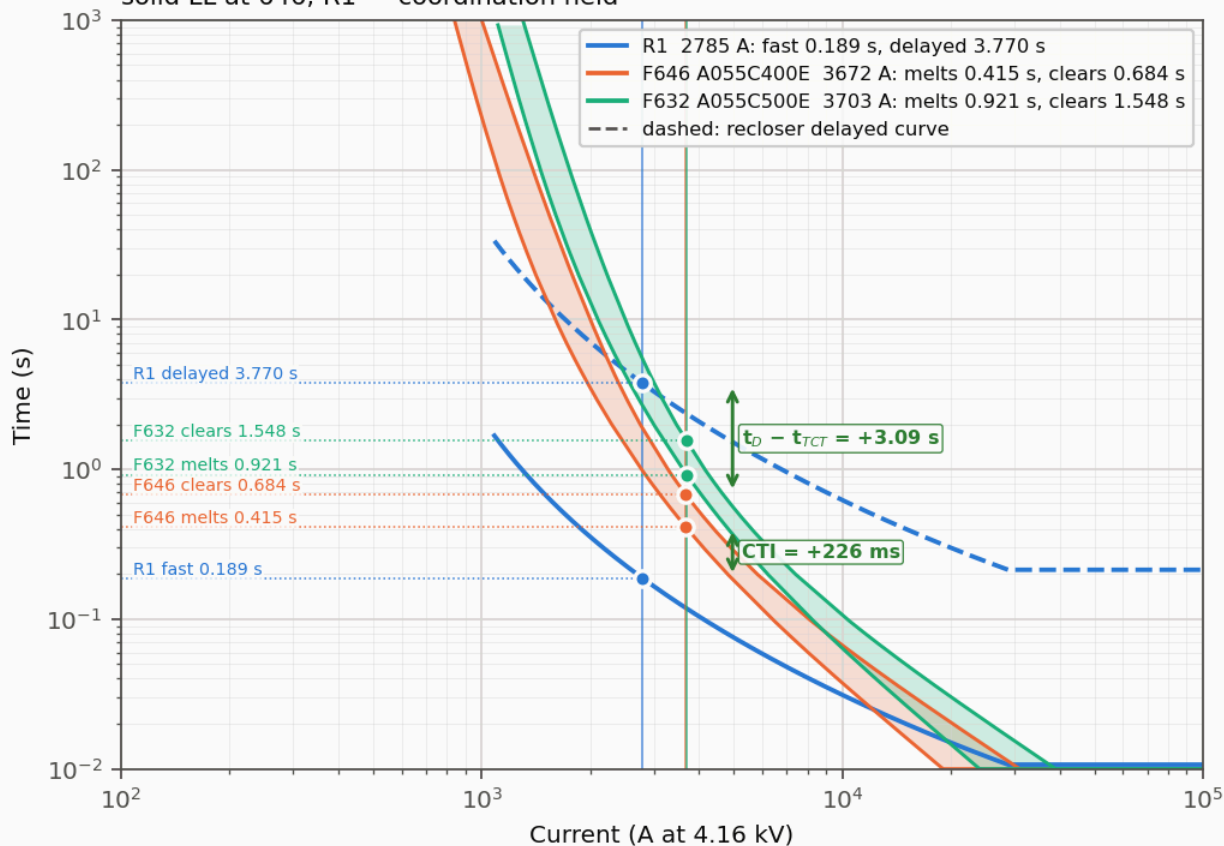


not applicable

# solid LL at 646, R2 as DSDR - coordination held



# solid LL at 646, R1 - coordination held



# Coordination with DSDR, DG 4.05 MVA at 692

|     |     |     |     |     |     |     |     |     |     |     |     |     |    |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|----|
| LG  |     |     | —   |     |     |     |     |     |     |     |     |     |    |
| LL  |     |     | —   |     |     |     |     |     |     |     | —   | —   |    |
| LLG |     |     | —   |     |     |     |     |     |     |     | —   | —   |    |
| LLL |     |     | —   | —   | —   |     |     |     |     | —   | —   | —   |    |
|     | 632 | 633 | 634 | 645 | 646 | 671 | 692 | 675 | 680 | 684 | 652 | 611 | DL |

Faulted node



coordination held



coordination lost



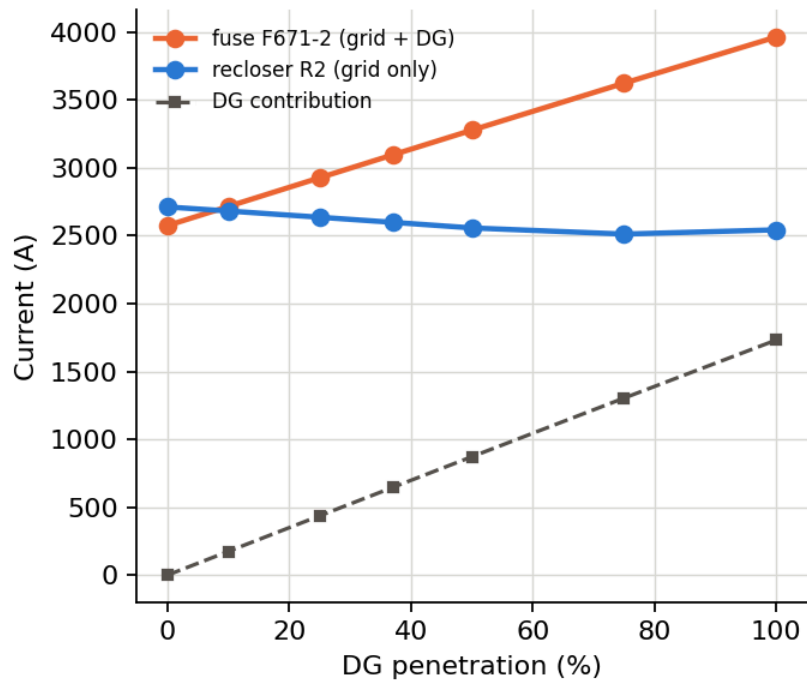
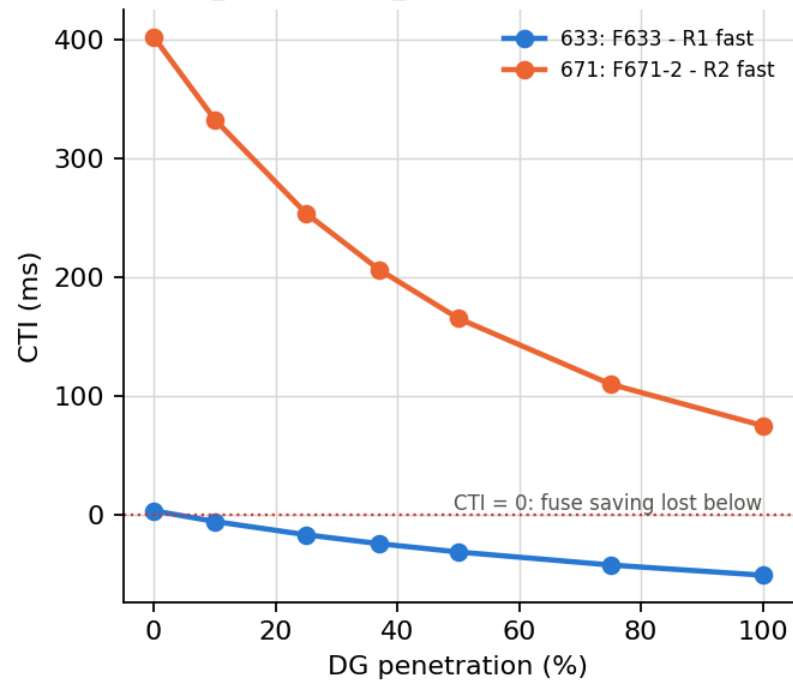
not applicable

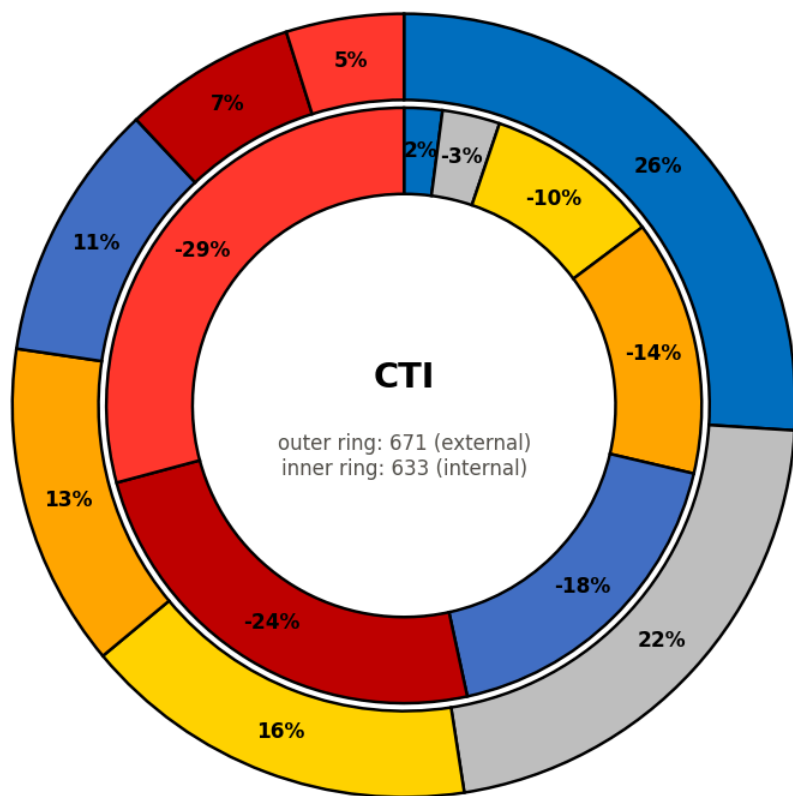


Fault LLL at 633: currents

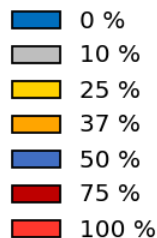


Fault LL at 684 (lateral from 671): currents

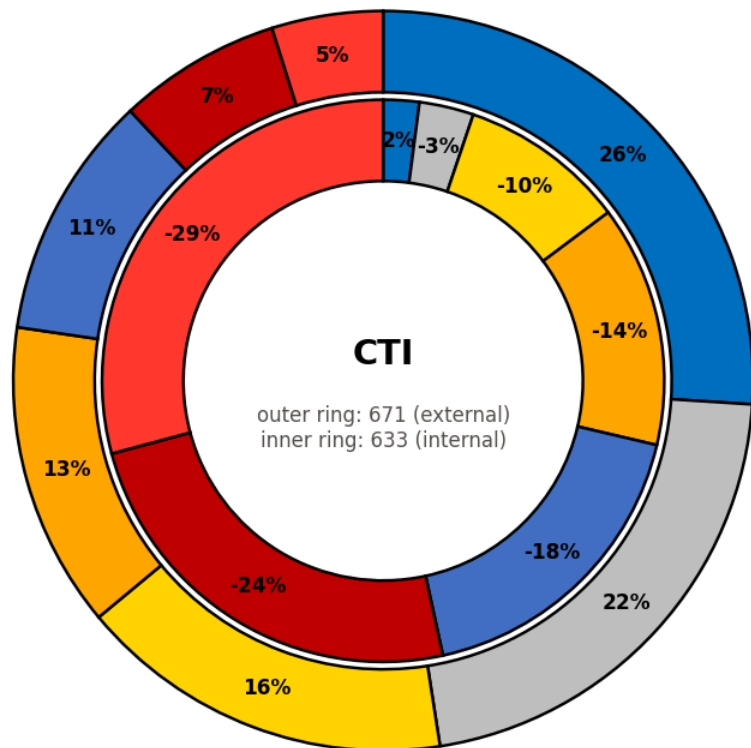
CTI =  $t_{\text{MMT}}(\text{fuse}) - t_{\text{fast}}(\text{recloser})$ 



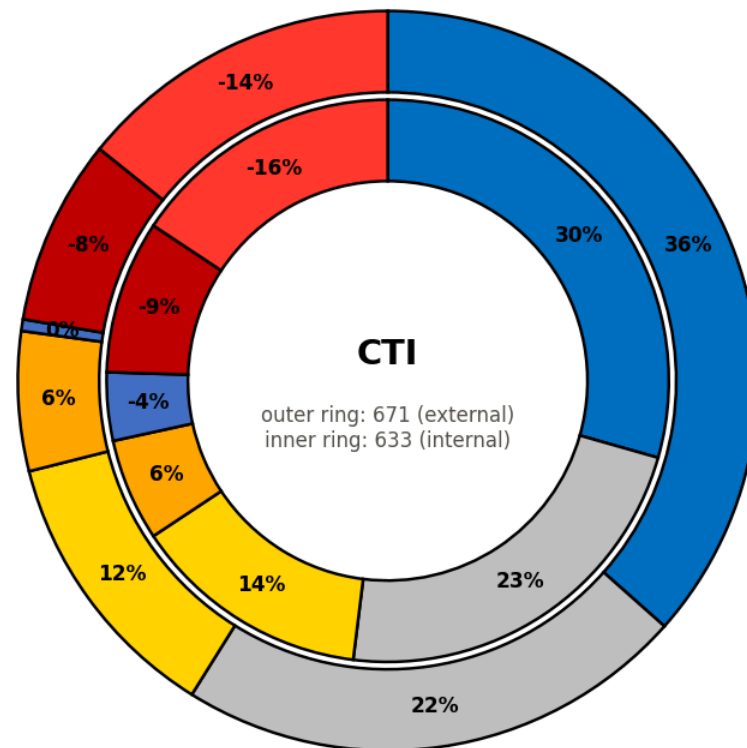
DG penetration



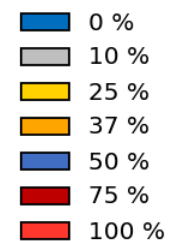
This study (PowerFactory)



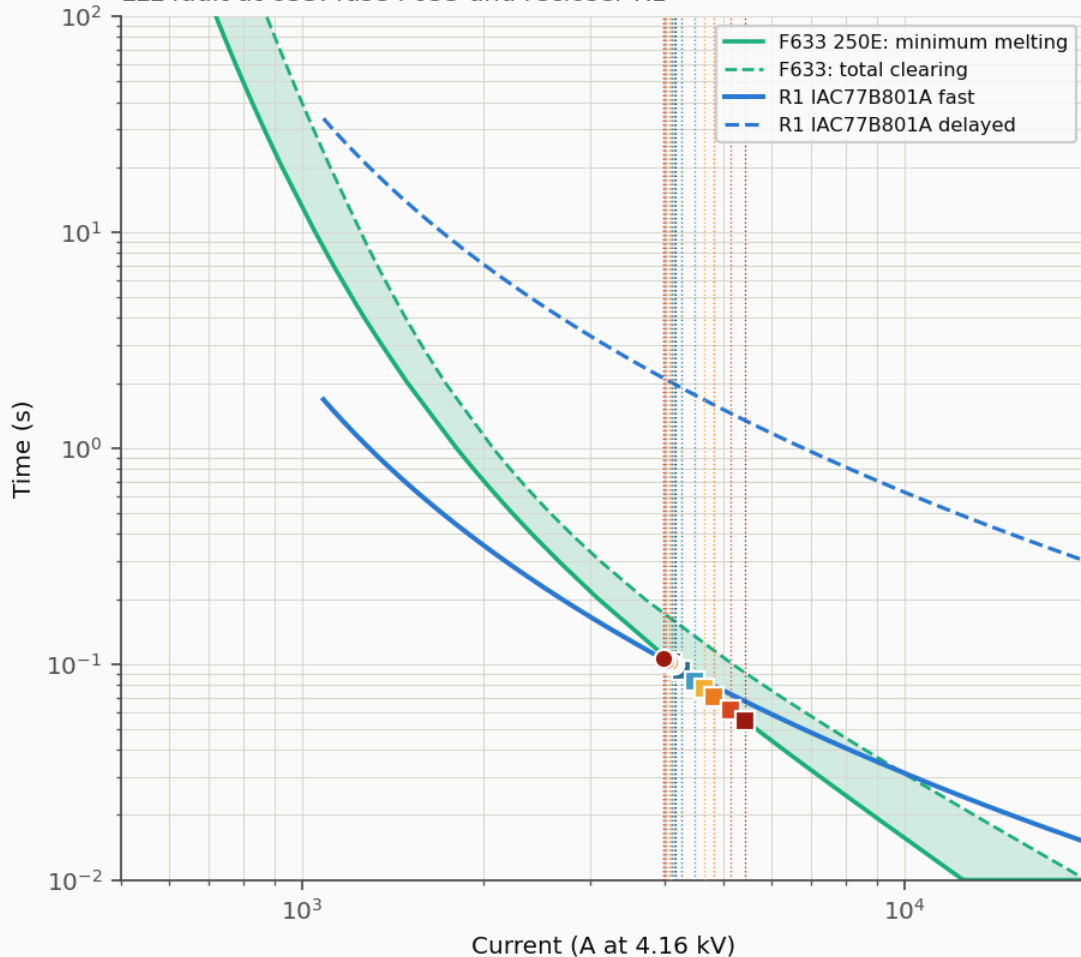
Reference paper (values read from its Fig. 7)



DG penetration



LLL fault at 633: fuse F633 and recloser R1



LL fault at 684 (lateral from 671): fuse F671-2 and recloser R2

